

Supplementary Material for A Hierarchical Approach with Crisis Mitigation for Multi-Robot Spatio-Temporal Restoration

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1 SUBROUTINES FOR ALGORITHM 1

In this appendix, we provide the detailed subroutines called by Algorithm 1 in the main paper.

Algorithm A PredictHighlyUrgentAreas($\tilde{k}, \tilde{\tau}, z$)

```

1:  $\Omega \leftarrow \emptyset, \hat{F} \leftarrow \emptyset, \hat{t} \leftarrow \emptyset$ 
2: for area  $j \in J$  do
3:    $\hat{t}_j \leftarrow t_j + \tilde{k} \cdot \tilde{\tau}(\neg j)$   $\triangleright$  Predicted time elapsed if not restored
4:    $\hat{F}_j \leftarrow g(\hat{t}_j, \delta_j)$   $\triangleright$  Predicted decay
5:   if  $\hat{F}_j \leq z$  then  $\triangleright$  Predicted highly urgent area
6:      $\Omega \leftarrow \Omega \cup \{j\}$ 
7:    $\hat{F} \leftarrow \hat{F} \cup \{\hat{F}_j\}; \hat{t} \leftarrow \hat{t} \cup \{\hat{t}_j\}$ 
8: return  $\Omega, \hat{F}, \hat{t}$ 

```

Algorithm B InternalOverload(C, Ω_C)

```

1: actual  $\leftarrow \{j \in C : F_j \leq z\}$   $\triangleright$  Actual highly urgent areas in  $C$ 
2: if  $|\Omega_C| > \frac{|C|}{2}$  and actual  $\neq \emptyset$  then  $\triangleright$  Internal overload
3:   return True
4: else
5:   return False

```

Algorithm C ExternalFeasibility($C, \hat{F}, \Omega_C$)

```

1: anchor  $\leftarrow \arg \min_{j \in C} \hat{F}_j$ 
2: feasible  $\leftarrow \emptyset$   $\triangleright$  set of feasible durations
3: for available  $r \in \mathbf{r}$  not assigned to  $C$  do  $\triangleright$  external robots
4:   dist  $\leftarrow \text{DISTANCE}(r, \text{anchor})$   $\triangleright$  distance robot from anchor
5:   duration  $\leftarrow \frac{\text{dist}}{v}$   $\triangleright$  travel duration given robot velocity
6:   if  $b_r \geq \frac{\text{duration}}{b_{\text{rate}}}$  then  $\triangleright$  battery can cover travel to anchor
7:      $\text{feasible} \leftarrow \text{feasible} \cup \{\text{duration}\}$ 
8: if feasible  $= \emptyset$  then
9:   return false
10:  $t_e \leftarrow \min(\text{feasible})$   $\triangleright$  robot can arrive
11: if  $\left\lfloor \frac{t_e}{\tilde{\tau}(C)} \right\rfloor < \min(|\Omega_C|, \left\lfloor \frac{b_r \cdot b_{\text{rate}}}{\tilde{\tau}(C)} \right\rfloor)$  then  $\triangleright$  robot can help
12:   return True
13: else
14:   return False

```

Algorithm D FormSurgeRobots($\mathcal{M}, \Gamma, \hat{t}$)

```

1:  $\mathbf{r}_b \leftarrow \{\text{home robot of } C \mid C \in \mathcal{M}\}$   $\triangleright$  corresponding home robots
2:  $\mathcal{P} \leftarrow \text{available robots } x \notin \mathbf{r}_b$   $\triangleright$  available external robots
3:  $\mathbf{r}_p \leftarrow \emptyset$ 
4:  $\mathbf{r}_{\text{surge}} \leftarrow \mathbf{r}_b$   $\triangleright$  initialize with home robots
5: if  $|\mathbf{r}_{\text{surge}}| < |\Gamma|$  then
6:    $PV_{\text{surge}} = \text{PVSURGERESPONSE}(\Gamma, \hat{t}, |\mathcal{P}|)$ 
7:   cost  $= \text{COSTSURGERESPONSE}(\Gamma, |\mathcal{P}|)$ 
8:   for  $x \in \mathcal{P}$  do
9:      $PV_{\text{home}} = \text{PVCLUSTERLOSS}(C_x, \hat{t})$ 
10:    if  $(PV_{\text{home}} < PV_{\text{surge}})$  and  $(b_x \geq \text{cost})$  then
11:       $\mathbf{r}_p = \mathbf{r}_p \cup \{x\}$   $\triangleright$  viable external robots
12: if  $\mathbf{r}_p \neq \emptyset$  then
13:    $\mathbf{r}_{\text{surge}} = \mathbf{r}_b \cup \mathbf{r}_p$   $\triangleright$  surge team
14: return  $\mathbf{r}_{\text{surge}}$ 

```

Algorithm E PVClusterLoss($C, \hat{t}$)

```

1: loss  $\leftarrow 0$ 
2: for area  $j \in C$  do
3:   loss  $= \text{loss} + \frac{1}{(1+\gamma)^{|C|}} \cdot L(\hat{t}_j, \delta_j)$   $\triangleright$  discounted losses
4: return loss

```

Algorithm F PVSurgeResponse($\Gamma, \hat{t}, |\mathcal{P}|$)

```

1:  $PV_{\Gamma} = \text{PVCLUSTERLOSS}(\Gamma, \hat{t})$ 
2: return  $\frac{PV_{\Gamma}}{\max(1, |\mathcal{P}|)}$ 

```

Algorithm G CostSurgeResponse($\Gamma, |\mathcal{P}|$)

```
1: size =  $\left\lfloor \frac{|\Gamma|}{\max(1, |\mathcal{P}|)} \right\rfloor$   $\triangleright$  expected size of surge response
2: duration  $\leftarrow 0$ 
3: for area  $j \in \Gamma$  do
4:   duration = duration +  $\bar{\tau}(j)$   $\triangleright$  travel duration
5: return size  $\cdot \left( \frac{\text{duration}}{|\Gamma|} \right) \cdot \left( \frac{1}{b_{\text{rate}}} \right)$ 
```

Algorithm H BindSurgeRobots($\mathbf{r}_{\text{surge}}$)

```
1: for robot  $r$  in  $\mathbf{r}_{\text{surge}}$  do
2:   Lock all areas  $j \in C_{(\text{surge}, r)}$  surge cluster
3:   Bind  $r \rightarrow C_{(\text{surge}, r)}$  with surge binding  $B_r$  (home cluster,
   areas, dwell cap, timestamp)
```

Algorithm I ReleaseSurgeRobots($\mathbf{r}_{\text{surge}}, \Omega$)

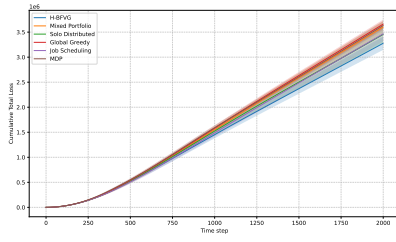
```
1: for robot  $r \in \mathbf{r}_{\text{surge}}$  with binding  $B_r$  do
2:    $C_{(\text{surge}, r)} \leftarrow$  surge cluster assigned to  $r$ 
3:   if not INTERNALOVERLOAD( $C_{(\text{surge}, r)}, \Omega$ ) then
4:     Unlock  $B_r.\text{areas}$ 
5:      $C_r \leftarrow$  home cluster of  $r$ 
6:      $\text{free} \leftarrow \{j \in C_r : j \text{ not locked by any surge robot}\}$ 
7:     Dispatch  $r$  back to  $C_r$  with areas  $\text{free}$ 
8:   Remove binding  $B_r$ 
```

2 ADDITIONAL EXPERIMENTS

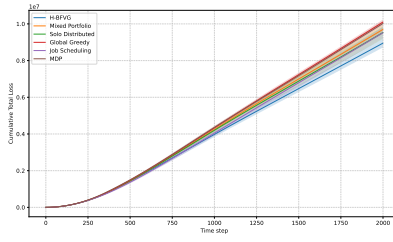
We evaluated the proposed approach also on a scenario without spatial correlation. The results (Fig. B) show that the proposed approach has the best performance, minimum cumulative losses. It also has the least duration that an area stays below the critical threshold (Table 1).

Baseline	4×24	8×64	10×100
Mixed portfolio	42% (p=0.00)	25% (p=0.00)	29% (p=0.00)
Solo distributed	25% (0.00)	23% (0.00)	29% (0.00)
Global greedy	49% (0.00)	41% (0.00)	37% (0.00)
Job scheduling	26% (0.00)	16% (0.00)	21% (0.00)
MDP	50% (0.00)	40% (0.00)	35% (0.00)

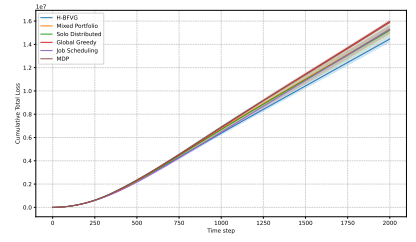
Table 1: (Without spatial correlation) Improvement of proposed approach in average duration areas are below critical threshold for m robots $\times n$ areas over baselines repeated 1,000 trials, with Welch one-tailed t-test for statistical significance



(a) Total losses: 4 robots $\times$ 24 areas

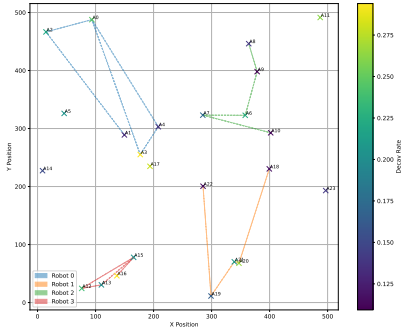


(b) 8 robots $\times$ 64 areas

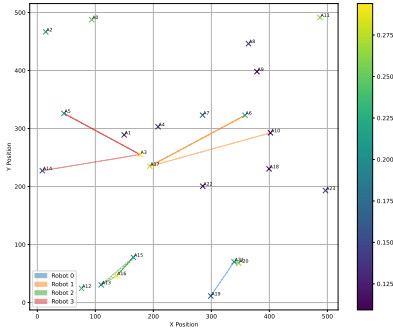


(c) 10 robots $\times$ 100 areas

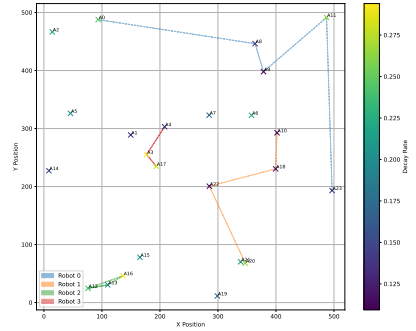
Figure A: (Without spatial correlation) Proposed H-BFVG has best overall performance over baselines, with least total losses across m robots $\times$ n areas repeated 1,000 trials.



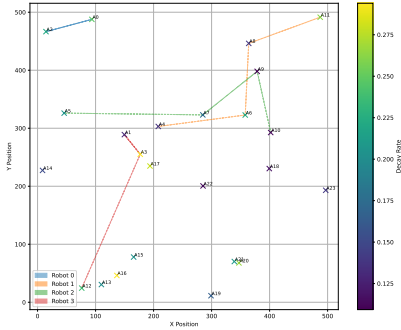
(a) H-BFVG



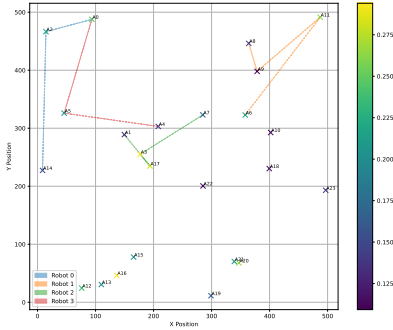
(b) Mixed Portfolio



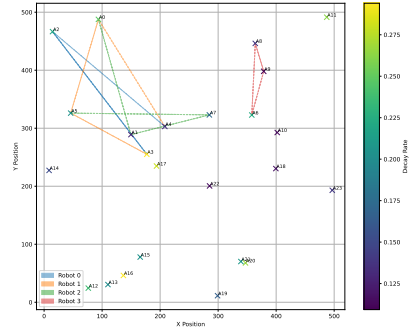
(c) Solo Distributed



(d) Global Greedy



(e) Job Scheduling



(f) MDP

Figure B: Routing behavior snapshot 4 robots $\times$ 24 areas w/o spatial correlation.